

CH14.2 矩陣的應用

$$1.(E) \quad 2. \begin{bmatrix} 1 & -1 \\ 1 & -2 \end{bmatrix} \quad 3. \begin{bmatrix} 6 & -2 \\ -7 & 9 \end{bmatrix} \quad 4. (1) \begin{bmatrix} \frac{1}{2} & 1 \\ -\frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$

$$(2) \begin{bmatrix} -1022 & -2046 \\ 1023 & 2047 \end{bmatrix} \quad 5. \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix} \quad 6. \begin{bmatrix} \frac{31}{2} & \frac{11}{2} \\ \frac{113}{2} & \frac{41}{2} \end{bmatrix} \quad 7.5$$

$$8.2 \quad 9. \frac{21}{64} \quad 10. (1) \begin{bmatrix} 0.2 & 0.6 \\ 0.8 & 0.4 \end{bmatrix} \quad (2) 0.52 \quad 11.9 \quad 12.35^\circ$$

$$13. 18x^2 - 18xy + 5y^2 = 18$$

解析

1. 沒有反方陣/行列式值為 0

故選(E)

$$2. A^3 = A^2 \cdot A$$

$$\therefore A = (A^2)^{-1} \cdot A^3 = \frac{1}{1} \begin{bmatrix} 3 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 2 & -5 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & -2 \end{bmatrix}$$

$$3. AX + 3B = C \Rightarrow AX = C - 3B$$

$$\begin{aligned} \Rightarrow X &= A^{-1}(C - 3B) = \frac{1}{-2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} \left(\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} - 3 \begin{bmatrix} 2 & -1 \\ 1 & -1 \end{bmatrix} \right) \\ &= -\frac{1}{2} \begin{bmatrix} 4 & -2 \\ -3 & -1 \end{bmatrix} \begin{bmatrix} -4 & 4 \\ -2 & 6 \end{bmatrix} = \begin{bmatrix} 6 & -2 \\ -7 & 9 \end{bmatrix} \end{aligned}$$

$$4. (1) \det P = \begin{vmatrix} -2 & -4 \\ 2 & 2 \end{vmatrix} = 4$$

$$P^{-1} = \frac{1}{4} \begin{bmatrix} 2 & 4 \\ -2 & -2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 1 \\ -\frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$

$$\begin{aligned} (2) B^{10} &= (PAP^{-1})^{10} = (PAP^{-1})(PAP^{-1}) \cdots (PAP^{-1}) \\ &= PA^{10}P^{-1} \end{aligned}$$

$$\begin{aligned}
&= \begin{bmatrix} -2 & -4 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 2^{10} & 0 \\ 0 & 1^{10} \end{bmatrix} \begin{bmatrix} \frac{1}{2} & 1 \\ -\frac{1}{2} & -\frac{1}{2} \end{bmatrix} \\
&= \begin{bmatrix} -2048 & -4 \\ 2048 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & 1 \\ -\frac{1}{2} & -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} -1022 & -2046 \\ 1023 & 2047 \end{bmatrix}
\end{aligned}$$

$$5. M \begin{bmatrix} 11 & 14 \\ -3 & -4 \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$\begin{aligned}
\Leftrightarrow M &= \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \begin{bmatrix} 11 & 14 \\ -3 & -4 \end{bmatrix}^{-1} = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \cdot \frac{1}{-44 + 42} \begin{bmatrix} -4 & -14 \\ 3 & 11 \end{bmatrix} \\
&= \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \begin{bmatrix} 2 & 7 \\ -\frac{3}{2} & -\frac{11}{2} \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}
\end{aligned}$$

$$\begin{aligned}
6. Y &= B^{-1}CA^{-1} = \left(\frac{1}{\begin{vmatrix} 4 & -1 \\ -3 & 1 \end{vmatrix}} \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix} \right) \begin{bmatrix} 5 & -1 \\ 4 & 2 \end{bmatrix} \left(\frac{1}{\begin{vmatrix} 2 & -1 \\ -4 & 3 \end{vmatrix}} \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix} \right) \\
&= \begin{bmatrix} 9 & 1 \\ 31 & 5 \end{bmatrix} \begin{bmatrix} \frac{3}{2} & \frac{1}{2} \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} \frac{31}{2} & \frac{11}{2} \\ \frac{113}{2} & \frac{41}{2} \end{bmatrix}
\end{aligned}$$

$$7. A = A^{-1} \Rightarrow AA = A^{-1}A \Rightarrow A^2 = I_2$$

$$\begin{bmatrix} 6 & -7 \\ k & k-11 \end{bmatrix} \begin{bmatrix} 6 & -7 \\ k & k-11 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{cases} 36 - 7k = 1 \Rightarrow k = 5 \\ -42 - 7(k - 11) = 0 \Rightarrow k = 5 \\ 6k + k(k - 11) = 0 \Rightarrow k = 5 \text{ 或 } 0 \\ -7k + (k - 11)^2 = 1 \Rightarrow k = 5 \text{ 或 } 24 \end{cases}, \text{故取 } k = \underline{5}$$

8. 設原坐標為 (x, y) , 線性變換後新坐標為 (x', y')

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\Rightarrow \frac{1}{5} \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\Rightarrow \begin{cases} x = \frac{1}{5}x' - \frac{2}{5}y' \\ y = \frac{1}{5}x' + \frac{3}{5}y' \end{cases}$$

代入 $x + y = 4$ 可得 $\left(\frac{1}{5}x' - \frac{2}{5}y'\right) + \left(\frac{1}{5}x' + \frac{3}{5}y'\right) = 4$

$$\Rightarrow 2x' + y' = 20$$

$\Leftrightarrow$ 新直線 $2x + y = 20$ 斜率為-2

$$\begin{array}{l} \text{一白一黑} < \begin{cases} \text{一白一黑} & \frac{3}{4} \\ \text{二白} & \frac{1}{4} \end{cases} \\ \text{二白} < \begin{cases} \text{一白一黑} & \frac{1}{2} \\ \text{二白} & \frac{1}{2} \end{cases} \end{array}$$

9. 甲袋情形有一白一黑，二白

轉移矩陣 $A = \begin{bmatrix} \frac{3}{4} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} \end{bmatrix}$

$$\begin{bmatrix} \frac{3}{4} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{3}{4} \\ \frac{1}{4} \end{bmatrix}, \begin{bmatrix} \frac{3}{4} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{3}{4} \\ \frac{1}{4} \end{bmatrix} = \begin{bmatrix} \frac{11}{16} \\ \frac{5}{16} \end{bmatrix}$$

$$\begin{bmatrix} \frac{3}{4} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{11}{16} \\ \frac{5}{16} \end{bmatrix} = \begin{bmatrix} \frac{43}{64} \\ \frac{21}{64} \end{bmatrix}$$

故甲袋有二白球之機率 $= \frac{21}{64}$

10. (1) 可得轉移矩陣 $P = \begin{bmatrix} 0.2 & 0.6 \\ 0.2 & 0.6 \\ 0.8 & 0.4 \end{bmatrix}$ 明晚上網

(2) 設 $X_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$\Rightarrow X_2 = PX_1 = \begin{bmatrix} 0.2 & 0.6 \\ 0.8 & 0.4 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0.8 \end{bmatrix}$$

$$\Rightarrow X_3 = PX_2 = \begin{bmatrix} 0.2 & 0.6 \\ 0.8 & 0.4 \end{bmatrix} \begin{bmatrix} 0.2 \\ 0.8 \end{bmatrix} = \begin{bmatrix} 0.52 \\ 0.48 \end{bmatrix}$$

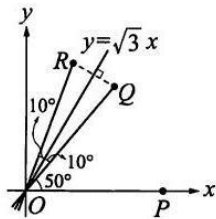
$\therefore$ 星期三也上網的機率為 0.52

11. $\overrightarrow{PQ} = (1, 1), \overrightarrow{PR} = (2, -4)$

$$\Delta PQR \text{ 面積} = \frac{1}{2} \left| \begin{vmatrix} 1 & 2 \\ 1 & -4 \end{vmatrix} \right| = 3$$

$$\text{又 } \det M = \det B \cdot \det A = \begin{vmatrix} \frac{4}{5} & -\frac{3}{5} \\ \frac{3}{5} & \frac{4}{5} \end{vmatrix} \cdot \begin{vmatrix} 1 & 0 \\ 2 & 3 \end{vmatrix} = 3$$

$$\text{故 } \Delta P'Q'R' \text{ 面積} = \Delta PQR \text{ 面積} \cdot \det M = 3 \cdot 3 = 9$$



12. 由題意令 P 旋轉 50° 到 Q ,

再對 $y = \sqrt{3}x$ 鏡射至 R

$$y = \sqrt{3}x \text{ 其斜率為 } \tan \theta' = \sqrt{3}$$

$$\Rightarrow \theta' = 60^\circ$$

$\angle POQ = 50^\circ$ 且 $\angle QOR$ 的角平分線

$$\text{為 } y = \sqrt{3}x$$

$$\therefore \angle QOR = 20^\circ \Rightarrow \angle POR = 70^\circ$$

由 P 到 R 即對 $y = \tan 35^\circ x$ 鏡射

$$\text{故 } \theta = 35^\circ$$